

Ayesha Adil

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Education

McMaster University – B.Eng. BME, Mechanical and Biomedical Engineering. Dean's Honor List. April 2026

- **Relevant Coursework:** Heat Transfer, Mechanics of Materials, Materials Science, Electrical Circuits, Measurements

Experience

Mechanical Design Engineer Co-op, Enbridge Gas – Chatham, ON May 2024 – Aug 2025

- Served as Mechanical Engineering Lead on an \$8M brownfield gas station expansion while supporting 3 concurrent projects, managing technical scope, schedule commitments, and multi-disciplinary deliverables.
- Reviewed contractor design packages (P&IDs, electrical line diagrams, engineering drawings, pipe sizing calculations) in Bluebeam Revu for regulatory compliance and alignment with Enbridge standards, redlining discrepancies as required.
- Researched applicable industry codes and standards (CSA Z662, MTO/TAC, Enbridge Engineering Standards) to support compliant mechanical design decisions and project documentation.
- Performed a gap analysis across 200+ legacy engineering standards and typical drawings following a company merger, identifying outdated and duplicate files and collaborating with SMEs to develop a standardized engineering masterlist.
- Supported the root cause investigation of a Pressure Relief Valve (PRV) field event involving an incorrectly installed U-bolt, identifying missing engineering drawing details and contributing to corrective design documentation.
- Developed a VBA tool to locate missing Material Test Reports (MTRs) across 200+ purchase orders, saving 50+ engineering hours, and authored a standardized filing procedure to prevent documentation gaps during project turnover.
- Structured data lists for maintenance teams to streamline asset uploads into CMMS platforms (Maximo, Oracle).

Mechanical Engineering Intern, ReFem Tech Startup – Toronto, ON Sept 2025 – June 2026

- Applied GD&T principles to define tolerances, clearances, and fits; created manufacturing drawings and revised designs.
- Met with manufacturers to discuss device requirements and design-for-manufacturing (DFM) principles.
- Reviewed CAD models with external manufacturers, updating part geometry to resolve early manufacturability concerns.
- Coordinated weekly project update meetings with 5 multidisciplinary team members, tracked milestones and prototype deadlines using Gantt charts, and followed up on project tasks and deliverables.

Projects

Biofill: Syringe-Filling Device (Patent Pending, Inventor) Capstone Project

- Developed an automated bioink syringe-filling device for Tessella Biosciences, adopted into their workflow to dispense 2–10 mL syringes (± 0.05 mL accuracy) in a biosafety cabinet using 95% 3D-printed components
- Designed the table-up drive system with a twist-and-lock piston attachment, sliding tray, PTFE bearing pads, and T-slotted steel rods; maintained the master CAD assembly across multiple design iterations.
- Applied GD&T to specify tolerances, fits, and clearances on engineering drawings while selecting standard fasteners and mechanical components to maintain a manufacturing BOM below \$1,500.
- Co-authored the invention disclosure with engineering drawings, BOM, spare parts list, technical specifications, design history, and testing data for manufacturing and patent submission

Extracurricular

McMaster Design League – CAD Designathons Jan 2023, 2024, 2025

- Placed 2nd (2023) and 1st (2025) in 24-hour SolidWorks designathons, presenting design solutions in teams of 4.

Food Bank Volunteer, Simcoe Hall, Oshawa June 2026 – Present

- Register clients weekly and prepare food packages for distribution.

Tax Volunteer, Canadian Volunteer Income Tax Program (CVITP) Sept 2025 – Present

- Filed 100+ tax returns for low-income clients at free community clinics with professionalism and confidentiality.

Skills

Technical: Excel (VBA), SolidWorks, Autodesk Inventor, GD&T, Bluebeam Revu, Maximo, Oracle, Power BI, Kahua